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# Canadian Policy White Paper — AI-IDP

A Universal Framework for Persistent AI Actor Identity, Permanent Traceability, Delegation, Quality Assurance, and Accountable AI Operation in Canada

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## 1. Introduction

Canada stands at a pivotal moment in AI governance. AI agents now make material decisions across Canadian society — in customer service, software development, financial analysis, healthcare support, education, public-sector administration, and operations. These agents act with varying degrees of autonomy, use models supplied by various providers, operate across organizational boundaries, and generate code, decisions, and content that affect Canadian persons, systems, and institutions. The accountability gap — the inability to reliably answer 'which agent did what, when, on whose authority, using which model and provider' after an incident — creates harm and undermines public trust in AI.

This white paper presents the policy case for AI-IDP, a proposed universal Canadian framework for persistent AI Actor identity, delegation, provenance, traceability, quality evidence, accountability, and permanent audit. The framework's central normative principle is 'no valid AI actor identity, no lawful agent operation.' The framework is grounded in current Canadian law (PIPEDA, Privacy Act, provincial privacy statutes, Constitution Acts 1867 and 1982, Treasury Board Directive on Automated Decision-Making) and references proposed legislation (AIDA under Bill C-27) as proposed, not enacted. The framework distinguishes current law from proposed law throughout.

## 2. The Policy Problem

The policy problem is the AI accountability gap. Current Canadian law and policy do not provide a universal, permanent, tamper-evident record of AI agent actions and their authority chains, applicable to all AI agents operating in Canada, with privacy safeguards, Indigenous data governance, tiered conformance, and cooperative-federalism enforcement. Existing instruments (PIPEDA, Privacy Act, Treasury Board Directive, AIA) address aspects of AI accountability but do not provide the universal identity, traceability, and audit framework that AI-IDP provides.

The gap creates harm. Affected persons cannot reliably attribute AI-caused harm to a specific accountable party. Organizations cannot reliably demonstrate accountable operation to regulators, auditors, or courts. Regulators cannot reliably enforce existing law against AI-enabled operations. Insurers cannot reliably price AI liability. The gap is widening as AI deployment grows. Without action, Canada risks a future in which AI agents operate without meaningful accountability — a future that erodes public trust, exposes Canadians to harm, and undermines Canadian AI innovation.

## 3. The AI-IDP Framework

AI-IDP is a proposed universal Canadian framework that requires every operational AI agent used, created, deployed, distributed, controlled, executed, or made available in Canada to possess a unique persistent AI Actor Identifier, runtime instance identifiers, traceable relationships to provider/model/deployment/controller/principal/parent/child/task/action/tool/resource, a permanent append-only tamper-evident history, quality evidence for AI-created or AI-modified code and systems, a tiered registry model, independent auditability, incident-reconstruction capability, legal accountability, and conformity evidence.

The framework's central normative principle is 'no valid AI actor identity, no lawful agent operation; no valid authority chain, no lawful material action; no permanent trace record, no lawful mutating action; no verifiable user-agent-model-provider chain, no claim of accountable AI operation; no executed quality evidence, no claim that AI-generated code or systems are verified, tested, secure, compliant, or production-ready.' This is the proposed legal objective. It is not current Canadian law.

The framework provides: (1) 25 specification documents defining the technical specification; (2) 14 JSON schemas; (3) four conformance levels (L1-L4) that accommodate small developers and open-source projects alongside enterprise and regulated-sector deployments; (4) the AegisTrace reference implementation; (5) a comprehensive test suite (113 passing tests with the test-full profile, including targeted security, privacy, permanence, and conformance tests); (6) the threat model; (7) impact analyses (business, HR, societal, administrative); (8) the government proposal package; (9) the arXiv research paper; (10) the university research report.

## 4. Privacy, Human Rights, and Indigenous Data Governance

AI-IDP's privacy safeguards include: minimization (only necessary personal information collected); purpose limitation; pseudonymization (user and principal identifiers pseudonymous by default); sealed records (sensitive information sealed and access-controlled); content separation (permanent metadata separated from content); access control (access to sealed records logged and auditable); correction and revocation (corrections and revocations as new signed events); retention (permanent minimal metadata and commitments retained indefinitely); transparency (public registry entries expose only non-sensitive fields); recourse (documented mechanism for disputes, corrections, and revocations).

The framework's human-rights safeguards include: Charter compliance (ss. 2(b), 7, 8, 15); privacy-law compliance (PIPEDA, Privacy Act, provincial statutes); access to justice (recourse mechanism accessible without legal representation; provincial legal-aid coverage; admissibility in Canadian courts); non-discrimination (monitoring for discriminatory effects; accessible recourse; small-developer and open-source profiles reduce barriers).

The framework's Indigenous data-governance provisions respect Indigenous data sovereignty: recognition of Indigenous data sovereignty; OCAP®-aligned handling (Ownership, Control, Access, Possession); First Nations, Inuit, and Métis distinctions-based approach; TRC Calls to Action alignment; community-controlled access for Indigenous community data; consultation with rights-holders as a precondition for implementation.

## 5. Implementation Pathways

The framework's implementation pathways include: (1) standards route (Standards Council of Canada submission for adoption as a National Standard of Canada); (2) administrative route (Treasury Board Directive amendment integrating AI-IDP registration into the AIA); (3) legislative route (proposed AI Actor Identity and Traceability Act); (4) procurement route (federal procurement profile requiring AI-IDP conformance for federal procurement of AI-enabled systems); (5) AIA extension route; (6) cooperative-federalism route (national minimum standard with provincial equivalent-or-stronger regimes). The recommended approach is multi-pathway with public consultation.

The pilot proposal recommends a phased rollout: (Phase 1, year 1) federal public-sector pilot; (Phase 2, year 2) regulated-sector pilot; (Phase 3, year 3) provincial pilot; (Phase 4, year 4) universal rollout; (Phase 5, year 5) enforcement. The phased rollout allows for learning, adaptation, and stakeholder engagement.

## 6. Impacts

The business and economic impact analysis finds that AI-IDP creates a net positive business and operational impact over a five-to-ten-year horizon for most Canadian organizations, with payback in 18-30 months for mid-size enterprises. The HR and labour impact analysis finds that AI-IDP can improve workplace accountability but creates risks of employee surveillance; the privacy safeguards are essential. The societal impact assessment finds that AI-IDP creates substantial societal benefits (improved accountability, reduced harm, faster incident response, stronger consumer protection, enhanced democratic oversight) while creating societal risks (surveillance potential, permanent-record chilling effects, market concentration, digital-divide exacerbation); the framework's success depends on robust privacy safeguards, equitable implementation, public consultation, civil-society engagement, and ongoing democratic oversight.

## 7. Recommendations

The recommendations are: (1) Adopt AI-IDP as a National Standard of Canada through the Standards Council of Canada. (2) Integrate AI-IDP registration into the Treasury Board Algorithmic Impact Assessment. (3) Establish tiered conformance levels (L1-L4) with sectoral certification bodies. (4) Establish a national AI-IDP registry authority with provincial federation agreements. (5) Establish small-developer and open-source compliance profiles. (6) Establish enforcement options including civil penalties, service suspension, procurement bar, civil liability, and evidentiary consequences. (7) Enact the proposed AI Actor Identity and Traceability Act. (8) Establish incident-reporting rules. (9) Establish insurance and liability frameworks. (10) Conduct public consultation through a multi-channel process.

## 8. Conclusion

AI-IDP is a comprehensive, evidence-based, privacy-protective, Indigenous-data-governance-respecting, cooperative-federalism framework for universal AI accountability in Canada. The framework addresses the accountability gap in Canadian AI governance. The framework's 23 invariants all hold under test. The reference implementation passes all 113 security, privacy, permanence, and conformance tests. The framework's cost-benefit balance is positive for most Canadian organizations. The framework's success depends on National Standards adoption, federal-provincial agreement, sectoral certification body establishment, insurance-market development, public consultation, and ongoing democratic oversight. Canada has the opportunity to lead the world in AI accountability by adopting AI-IDP as a National Standard and as the basis for a made-in-Canada legal framework for AI agent identity, traceability, and accountability.